DSC-502: Data Analytics - Unit 1

Section A: Short Answer Questions

#	Question	Bloom's Level
1	Differentiate between univariate, bivariate, and multivariate analysis. Give one suitable example of each.	K2
2	What is the Curse of Dimensionality? Mention any two major consequences of high-dimensional data.	K2
3	What is the Kernel Trick in machine learning? Explain why it is useful for analyzing non-linear data.	K2
4	Differentiate between Feature Selection and Feature Extraction with one example of each.	K2
5	Define Degree Centrality and Betweenness Centrality in graph data analysis.	K1
6	Why is t-SNE primarily used for visualization rather than as a general-purpose dimensionality reduction technique?	K2
7	Differentiate between Mean, Median, and Mode. Under what circumstances is the Median preferred over the Mean?	K2
8	Define the Interquartile Range (IQR). Explain how the IQR method can be used to identify outliers.	K2
9	Define Pearson's correlation coefficient. Interpret the values $r=+1$, $r=-1$, and $r=0$.	K2
10	Write the equation of Simple Linear Regression and define each term in the equation.	K1
11	What is a Graph? Define Degree and Clustering Coefficient for a vertex.	K1
12	Define the Graph Laplacian Matrix. What is its significance in graph data analysis?	K2
13	Differentiate between PCA and LDA with respect to their objectives and whether they are supervised or unsupervised techniques.	K2
14	What is Sparsity in high-dimensional data? Why can sparse data create challenges for data analysis?	K2
15	Define Variance and Standard Deviation. What does a larger standard deviation indicate about a dataset?	K1

Section B: Long Answer Questions

#	Question	Bloom's Level
1	Explain the Curse of Dimensionality in detail. Discuss its major consequences with suitable examples. Explain how dimensionality reduction techniques such as PCA and t-SNE can help address problems associated with high-dimensional data.	K4
2	Explain Principal Component Analysis (PCA) in detail. Describe the complete procedure for performing PCA on a dataset, including standardization, covariance/correlation matrix construction, eigenvalue and eigenvector computation, selection of principal components, and transformation of the data. Explain the role of eigenvalues and eigenvectors in PCA.	K4
3	Compare and contrast PCA, LDA, t-SNE, and UMAP with respect to their objectives, supervised/unsupervised nature, preservation of information, computational characteristics, and applications. Explain when you would prefer each technique.	K5
4	Explain the major centrality measures used in graph data analysis, namely Degree Centrality, Betweenness Centrality, Closeness Centrality, and PageRank. Illustrate each measure using a suitable graph or real-world example.	K4
5	Explain kernel methods for non-linear data analysis. Describe the Kernel Trick and explain the Radial Basis Function (RBF) kernel, including its mathematical expression and important properties. Give a real-world scenario where an RBF kernel would be preferred over a linear kernel.	K4
6	Explain Univariate, Bivariate, and Multivariate Analysis in detail. Discuss their objectives, commonly used statistical measures, and appropriate visualizations with suitable examples.	K4
7	Derive Pearson's correlation coefficient and explain the meaning of each component in the formula. Using a suitable numerical dataset, calculate the correlation coefficient and interpret the result.	K4
8	Explain the concept of Multivariate Analysis. Discuss important techniques used for analyzing multiple variables, such as multiple regression, correlation analysis, PCA, and cluster analysis. Give a real-world example where multivariate analysis is more appropriate than univariate or bivariate analysis.	K4
9	Explain the working principle of t-SNE in detail. Discuss the concepts of high-dimensional and low-dimensional similarity, probability distributions, KL divergence, and perplexity. Explain how the choice of perplexity can affect the resulting visualization.	K4
10	Compare Feature Selection and Feature Extraction in detail. Discuss their objectives, techniques, advantages, disadvantages, and suitable examples. Explain when one approach may be preferred over the other.	K5

Section C: Numerical / Problem-Based Questions

#	Question	Bloom's Level
1	For the dataset [10, 12, 23, 23, 25, 28, 30, 35], calculate the Mean, Median, Mode, Population Variance, Population Standard Deviation, and IQR. Show all intermediate calculations.	K3
2	Calculate Pearson's correlation coefficient for the following data: X=[1,2,3,4,5] and Y=[2,4,5,7,8]. Interpret the strength and direction of the correlation.	K3
3	Fit a Simple Linear Regression model for the data points (1,2),(2,4),(3,5),(4,7),(5,8). Determine the regression equation and predict the value of Y when X=6.	K3
4	For the dataset [5,7,8,10,12,15,18,20,22,25,28,30], calculate Q1, Q3, and IQR. Using the 1.5×IQR rule, determine whether any observations are outliers.	K3
5	Consider the following undirected graph with vertices A,B,C,D,E and edges AB,AC,BC,CD,DE. Calculate the Degree Centrality, normalized Betweenness Centrality, and Local Clustering Coefficient for each vertex. Show all intermediate calculations.	K3